QOSAIN BUKHARI

Junior Machine Learning Engineer

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SUMMARY

Software Engineering graduate (BS, 2026) with a 3-month ML internship and five end-to-end projects across tabular ML, computer vision, NLP and Retrieval-Augmented Generation. Built and deployed a hotel cancellation predictor with a leakage-free, time-based evaluation and a live demo, and a full-stack accident-detection system around a custom YOLOv12 model (89.9% mAP@50 on a held-out test set). Comfortable across the ML workflow: data validation, honest evaluation, FastAPI serving, and reproducible pipelines with Docker, MLflow, DVC and GitHub Actions CI. Seeking a junior ML engineer or ML internship role.

EDUCATION

Mirpur University of Science & Technology (MUST) | 2022 – 2026
BS Software Engineering | CGPA: 3.40/4.0

EXPERIENCE

Machine Learning Intern | Pixels Inn Institute, Mirpur, Azad Kashmir | Jul 2025 – Oct 2025

- Prepared datasets with Pandas and NumPy (missing values, duplicates, encoding, scaling), fitting transforms on training data only to prevent leakage.
- Benchmarked linear, tree-based, and ensemble models in Scikit-learn using F1, ROC-AUC, MAE, and RMSE, tuning with cross-validation.
- Trained dense networks and a CNN in TensorFlow and PyTorch; built an OpenCV + YOLO detection workflow for images and video.
- Evaluated as “Excellent” by the Managing Director on tool use, design ability, ethics, communication, and motivation.

PROJECTS

Cancellation Radar: Hotel Booking Cancellation Prediction | 2026 | Live demo | GitHub
Python, XGBoost, Optuna, SHAP, MLflow Model Registry, DVC, Pandera, Evidently, FastAPI, Docker, Vercel, GitHub Actions

- Predicts cancellation risk at booking time on 119,390 bookings; found and removed 6 leaking check-in columns and tested on the newest 20% of arrivals (23,878 bookings) instead of a random split, giving an honest ROC-AUC of 0.824.
- Chose the decision threshold by business cost instead of 0.5: catches 94% of cancellations and cuts the cost of wrong decisions from €40.86 to €9.69 per booking (–76%) versus no model.
- Serves SHAP explanations, batch scoring and drift checks (PSI, Evidently) from a FastAPI app on Vercel; the MLflow registry promotes a retrained model only if it beats production.

AI Accident Detection & Emergency Response System (Final Year Project) | 2025 – 2026 | GitHub
Python, YOLOv12, OpenCV, FastAPI, React, MySQL, Alembic, Docker Compose, Pytest

- Trained a custom YOLOv12 model to detect road accidents in images and video across 3 classes (fire, moderate, severe): 89.9% mAP@50, 92.5% precision and 86.8% recall on a 1,800-image held-out test set.
- Built the complete system: FastAPI backend with JWT role-based authentication and MySQL, background video jobs, email/WhatsApp alerts with annotated frames, and a React dashboard; packaged with Docker Compose and covered with Pytest.

Document Q&A RAG Chatbot | 2026 | GitHub
Python, LangChain, ChromaDB, Sentence Transformers, Llama 3.1 (Groq), FastAPI, Streamlit, Docker

- Built a PDF Q&A chatbot with a RAG pipeline: chunking, MiniLM embeddings, persistent ChromaDB retrieval, Llama 3.1 generation, conversation memory, and page-level source citations.
- Manually evaluated on 20 Q&A pairs: 85% retrieval accuracy, 90% answer faithfulness, 3.2 s average response time. Containerized with Docker Compose; Pytest suite runs on GitHub Actions CI.

Fake News Detection: End-to-End MLOps Pipeline | 2025 | GitHub
Python, Scikit-learn, NLTK, TF-IDF, FastAPI, MLflow, DVC, Docker

- Built a reproducible pipeline with DVC-versioned stages and MLflow tracking, served through a FastAPI /predict endpoint in Docker; compared 4 classifiers on 72,134 articles, best F1 95.80% (TF-IDF + Random Forest).

Chest CT-Scan Classification: Deep Learning + MLOps | 2025 | GitHub
Python, TensorFlow, Keras, VGG16, Flask, MLflow, DVC

- VGG16 transfer-learning classifier in a 4-stage DVC pipeline with MLflow tracking; found and fixed a train/evaluation split mismatch that leaked a third of the evaluation images into training and produced a false 100% accuracy.

TECHNICAL SKILLS

Languages: Python, SQL
ML / Deep Learning: Scikit-learn, XGBoost, Optuna, TensorFlow, Keras, PyTorch, OpenCV, YOLOv12, CNNs, SHAP, Model Evaluation
GenAI / NLP: RAG, LLM Applications, LangChain, Llama 3, ChromaDB, Embeddings, Prompt Engineering, NLTK
MLOps / Backend: FastAPI, Flask, REST APIs, Docker, MLflow, DVC, Evidently, GitHub Actions (CI), Git, Pytest, MySQL
Data: Pandas, NumPy, Pandera, EDA, Feature Engineering, Leakage Analysis

CERTIFICATIONS

Deep Learning Specialization, DeepLearning.AI (2025) | Generative AI, CampusX (2026)
Udemy Data Science, ML, DL & NLP Bootcamp (2024) | Microsoft ML Concepts (2024)